CLASS 10 MATHEMATICS :: FA-1

Generating New Problems

Student Name: ..... **Roll No:** ..... **Class & Sec:** .....

1. Find the HCF and LCM of 72 and 120 using the prime factorisation method.

2. Check whether the rational number $23 / (2^3 \times 5^2)$ has a terminating decimal expansion, and write its decimal form without actual division.

3. Prove that $3 + 2\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is an irrational number.

4. Expand the expression $\log_{10}(343 / 64)$ using the standard laws of logarithms.

5. If $\log_2 x = 5$, determine the numerical value of x.

6. Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8\}$. Find $A \cup B$ and $A \cap B$.

7. If $A = \{x : x \text{ is a prime number less than } 10\}$ and $B = \{x : x \text{ is an odd positive integer less than } 10\}$, evaluate $A - B$ and $B - A$.

8. In a surveyed group of students, $n(A) = 25$, $n(B) = 18$, and $n(A \cap B) = 8$. Calculate $n(A \cup B)$ using the set formula.

9. Find the zeroes of the quadratic polynomial $p(x) = x^2 - 5x + 6$ and state their relation with coefficients.

10. Form a quadratic polynomial whose sum and product of zeroes are -4 and 3 respectively.